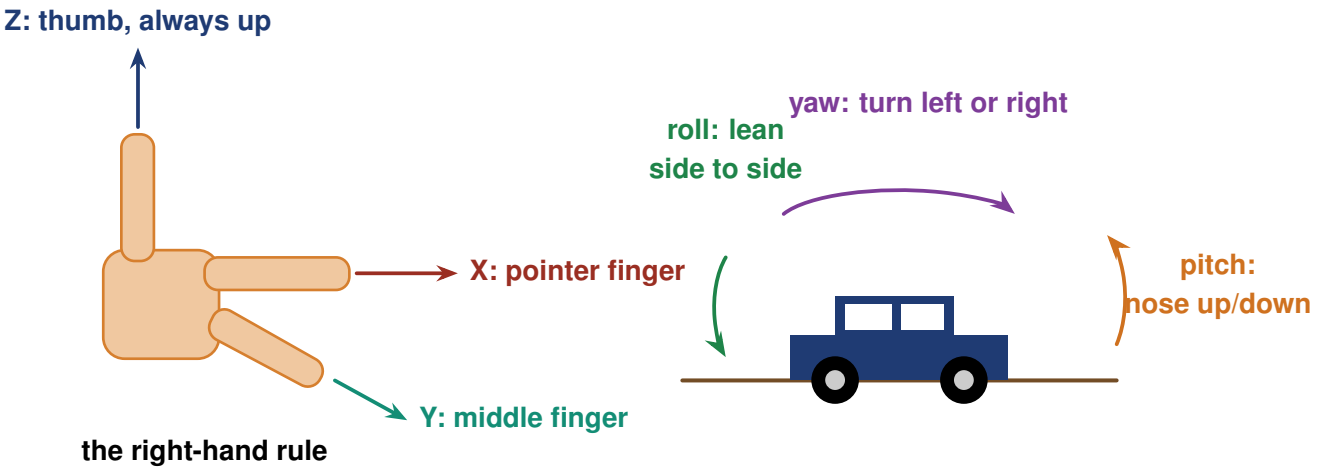
Rotation, Pose and Frames

BIG IDEA

Pose means where you are and which way you are facing.



THE IDEA

Stand in the middle of a room. Where you are is your spot on the floor. Which way you face is whether you look at the board or the window, and whether you tilt or lean. A LiDAR on a car needs to know both, all the time, because the car bumps, turns and leans. Together they are called its **pose**.

REMEMBER

- ✓ Pose = where you are + which way you are facing.
- ✓ Right hand: pointer finger is X, middle finger is Y, and the thumb is always Z.
- ✓ Roll is leaning, pitch is nose up or down, and yaw is turning left or right.

TRY THIS

Hold up your right hand: pointer finger forward, middle finger to the left, thumb up. Now turn your hand to show roll, pitch and yaw.

The full module is in the Reference Book.